Answer **One** question from this section

7 A uniform wooden rod of weight 50 N and length 1.0 m is gently lowered into water. The upper end of the rod is attached to a light string. When the rod is in equilibrium, the string is vertical and exactly half of the rod is underwater as shown in Fig. 7.1. The rod makes an angle θ with the surface of the water.

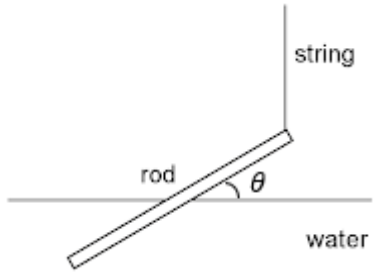

Fig. 7.1

(a) **(i)** Explain why the string is vertical.

……

…… [1]

(ii) On Fig. 7.1, draw the three forces tension T, upthrust U and the weight W acting on the rod. [2]

(iii) By considering moments about the centre of gravity of the rod, show that $T = 0.5\ U$.

[2]

(iv) Calculate the magnitude of T.

T = ……..N [2]

(v) By balancing the forces, determine the density of the wooden rod. The density of water is 1.0×10^3 kg m^{-3}.

density of rod = ...kg m^{-3} [4]

(b) **(i)** State the principle of the conservation of momentum.

.....

..... [1]

(ii) Object **A** of mass 1.2 kg collides head-on and elastically with object **B** of mass 0.60 kg moving with a speed 0.20 m s^{-1} towards it as shown in Fig. 7.2.

Fig. 7.2

After the collision, object **B** moves off with a speed of 0.10 m s^{-1} opposite to its initial motion.

Calculate the initial speed of object **A**.

initial speed of **A** = ... m s^{-1} [3]

[Turn over

(c) An 80 kg astronaut is at a distance of 30 m away from a space shuttle. He wishes to return to the space shuttle by means of a thruster. The thruster is attached to the body of the astronaut and it emits a stream of gas when it is turned on.

(i) State and explain the direction in which the gas has to be ejected for him to return to the space shuttle.

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.....

..... [3]

(ii) The gas is ejected at a constant rate of 0.5 kg s^{-1} and at a speed of 20 m s^{-1} relative to the astronaut for a period of 1.0 s.

Calculate the speed of the astronaut at the end of 1.0 s.

speed = ...m s^{-1} [2]